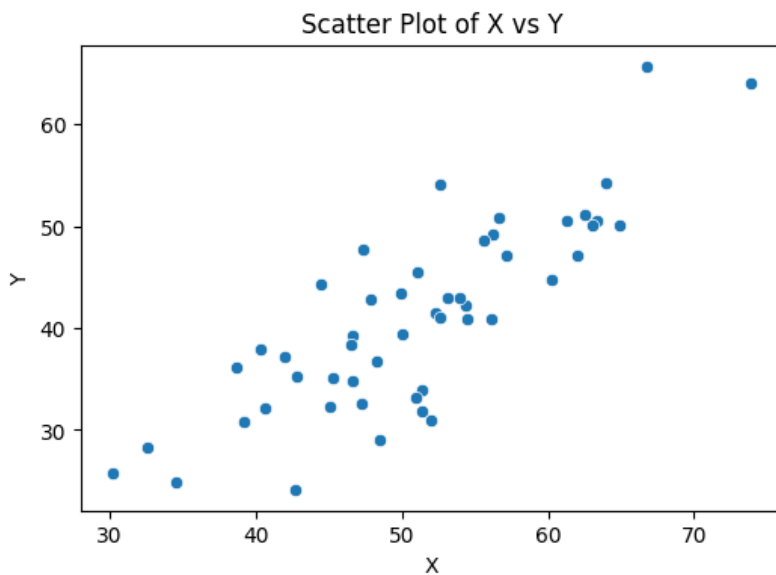


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

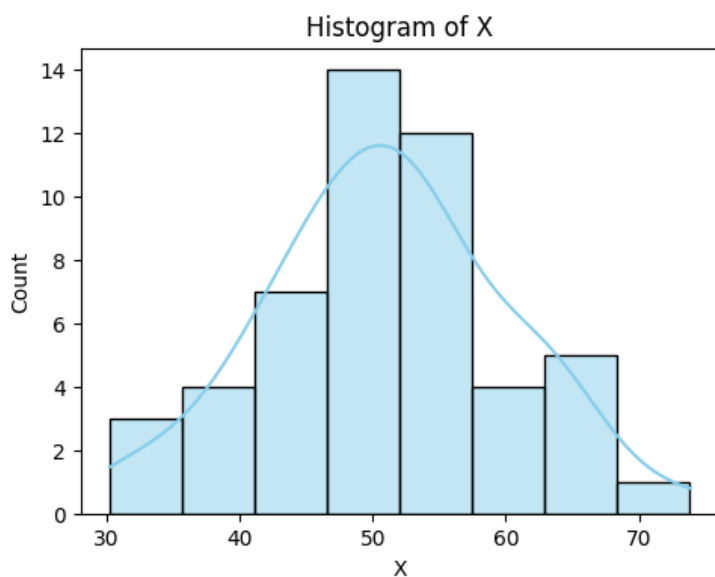
```
np.random.seed(10)
x = np.random.normal(50, 10, 50) # Normally distributed data
y = x * 0.8 + np.random.normal(0, 5, 50) # Slightly correlated data
data = pd.DataFrame({'X': x, 'Y': y})
```

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='X', y='Y', data=data)
plt.title("Scatter Plot of X vs Y")
plt.show()
```



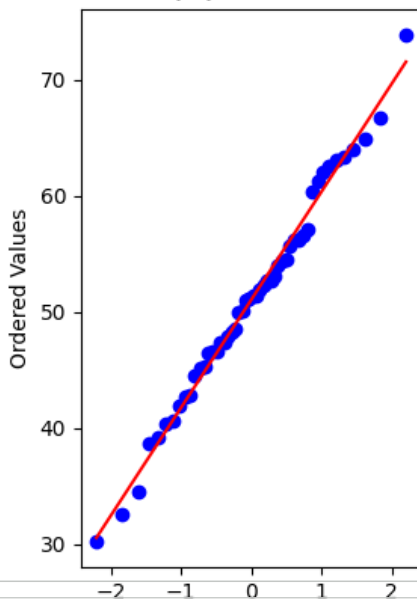
```
plt.figure(figsize=(12,4))
plt.subplot(1,2,1)
sns.histplot(data['X'], kde=True, color='skyblue')
plt.title("Histogram of X")
```

Text(0.5, 1.0, 'Histogram of X')



```
plt.subplot(1,2,2)
stats.probplot(data['X'], dist="norm", plot=plt)
plt.title("Q-Q Plot for X")
plt.show()
```

Q-Q Plot for X



```
shapiro_test = stats.shapiro(data['X'])
print("Shapiro-Wilk Test for X:")
print(f"Statistic = {shapiro_test.statistic:.4f}, p-value = {shapiro_test.pvalue:.4f}")
```

Shapiro-Wilk Test for X:
Statistic = 0.9919, p-value = 0.9800

```
skew_value = stats.skew(data['X'])
print(f"\nSkewness of X: {skew_value:.4f}")
if abs(skew_value) < 0.5:
    print("→ The data is approximately symmetric.")
else:
    print("→ The data is asymmetric.")
```

Skewness of X: 0.0313
→ The data is approximately symmetric.

```
spearman_corr, p_corr = stats.spearmanr(data['X'], data['Y'])
print(f"\nSpearman's Rank Correlation = {spearman_corr:.4f}")
print(f"p-value = {p_corr:.4f}")
if p_corr < 0.05:
    print("→ There is a significant correlation between X and Y.")
else:
    print("→ No significant correlation found.")
```

Spearman's Rank Correlation = 0.8012
p-value = 0.0000
→ There is a significant correlation between X and Y.